CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 3: Pair of Linear Equations in Two Variables

Every formula and result of the chapter on one sheet

1	General form of the pair	$a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$
2	Unique solution (intersecting lines)	$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
3	No solution (parallel lines)	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
4	Infinitely many solutions (coincident lines)	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
5	Consistent pair	Has at least one solution (intersecting or coincident lines)
6	Inconsistent pair	Has no solution (parallel lines)
7	Two-digit number	$10x + y$; with digits reversed, $10y + x$
8	Fixed charge problems	Total $= x + n\,y$, with fixed part x and rate y per unit
9	Speed, distance and time	$\text{distance} = \text{speed} \times \text{time}$
10	Line parallel to the axes	$x = a$ is parallel to the y-axis; $y = b$ is parallel to the x-axis